SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE CODE: CSA15

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Total Marks: 50

Dave's Taxonomy: Manipulation (Level 2) Question
Count: 5
Duration : 60 Minutes

Code of Conduct

I R.YUVASHREE(192521485) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R.YUVASHREE

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces. 3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and

propose a secure identity and privacy design using federation concepts.

Equivalent Steps in VMware Workstation Pro

- Library pane showing multiple VMs = multiple instances/regions.
- VM → Snapshot → Take Snapshot, then Snapshot Manager = backup/versioning demonstration.
- Edit → Virtual Network Editor to show NAT/Host-only/Bridged networks = hybrid or multi-cloud connectivity.
- VM → Manage → Clone = replica/auto-scaling demonstration.

Tip: Paste each screenshot directly under the relevant question in this document (Insert → Picture in Word) before final submission, with a one-line caption explaining what it demonstrates.

Analytical Problem Solving Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 – Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1. Compare Type-1 and Type-2 Hypervisors for Mission-Critical Applications

A hypervisor is software that creates and manages virtual machines (VMs), allowing multiple operating systems to share the resources of a physical computer.

Feature	Type-1 Hypervisor	Type-2 Hypervisor
Working	Runs directly on physical hardware.	Runs on top of a host operating system.
Performance	Generally offers higher performance with less host-OS overhead.	May have additional overhead because it depends on the host OS.
Security	Has a smaller software stack when properly configured, though it still requires patching and protection.	Security depends on both the hypervisor and the host operating system.
Manageability	Supports enterprise management, monitoring, and automation.	Convenient for individual desktop use and testing.
Examples	VMware ESXi, Xen, Microsoft Hyper-V.	VMware Workstation Pro, Oracle VirtualBox.
Typical use	Cloud datacenters and enterprise workloads.	Development, training, and local testing.

Analysis: Type-1 hypervisors access hardware directly and are designed for server environments. They can provide efficient resource allocation, centralized administration, and high availability when used with suitable datacenter tools. Type-2 hypervisors are easier to use on a personal computer but add dependence on the host operating system.

Conclusion: A Type-1 hypervisor is the better choice for mission-critical applications because it generally provides better performance, strong isolation capabilities, and enterprise-level management.

Security and availability still depend on correct configuration, updates, monitoring, and backup planning.

2. Workload Isolation in a Cloud Provider

A cloud provider may host finance, healthcare, and student portal workloads on the same physical server. Virtualization enables these workloads to share hardware while remaining logically separated.

Virtualization architecture:

- Physical server: supplies CPU, memory, storage, and network interfaces.
- Hypervisor: allocates hardware resources and separates virtual machines.
- Finance VM: runs financial applications and stores financial records.
- Healthcare VM: runs healthcare applications and protects patient records.
- Student VM: runs academic services and stores student information.

How isolation is supported:

- CPU and memory isolation: the hypervisor controls each VM's access to processor time and memory.
- Storage isolation: virtual disks, access permissions, and encryption help prevent unauthorized access to another workload's data.
- Network isolation: VLANs, virtual networks, firewalls, and security groups restrict communication between workloads.
- Identity and access control: separate accounts, roles, and permissions limit access to sensitive resources.

Two major attack surfaces:

1. 1. Hypervisor attack surface: A vulnerability in the hypervisor may allow an attacker to escape a VM and affect the host or other VMs.
2. 2. Virtual network attack surface: Misconfigured virtual networks, exposed services, or weak access controls may allow unauthorized communication or data theft.

Conclusion: Strong hypervisor security, network segmentation, encryption, least-privilege access, and regular patching are essential for safely hosting multiple tenants on shared hardware.

3. VM Host Utilization Analysis and Corrective Actions

The given utilization snapshot indicates that the host is under considerable pressure. CPU and memory usage are high, storage latency may be affecting performance, and VM startup is taking longer than desirable.

Metric	Observed value	Likely meaning
CPU utilization	92%	Very high usage; limited processing capacity remains.
Memory utilization	88%	High memory usage; memory pressure or swapping may occur.
Storage I/O latency	35 ms	Potential storage bottleneck, especially for latency-sensitive workloads.
Average VM boot time	170 seconds	Slow startup; may be caused by resource contention, storage delays, or startup services.

Likely bottlenecks

- CPU bottleneck: High CPU utilization can cause scheduling delays and reduce application responsiveness.
- Memory bottleneck: High memory usage can lead to memory reclamation, ballooning, or swapping, all of which may reduce performance.
- Storage bottleneck: A latency of 35 ms may indicate overloaded disks, high I/O queue depth, slow storage media, or a saturated storage network.
- VM boot delay: The 170-second boot time may result from CPU contention, memory pressure, slow disk reads, or many VMs starting simultaneously.

Recommended corrective actions

- CPU: Move some VMs to less-loaded hosts, balance workloads, and add CPU capacity if monitoring confirms sustained demand.
- Memory: Identify memory-heavy VMs, remove unnecessary allocations, and add RAM if required. Check whether swapping or ballooning is occurring.
- Storage: Check IOPS, queue depth, throughput, and storage-network performance. Consider faster SSD or NVMe storage if appropriate.
- Boot time: Stagger VM startup, investigate startup services, and reduce CPU and storage contention.
- Monitoring: Track utilization and latency over time and configure alerts before resources reach critical levels.

Conclusion: CPU and memory are under heavy pressure, while storage latency is a possible major performance bottleneck. Administrators should verify the cause using monitoring data, address storage and scheduling issues, and then redistribute or expand resources as needed.

4. How Software-Defined Datacenter (SDDC) Improves Agility

A Software-Defined Datacenter (SDDC) uses software to virtualize and centrally manage compute, storage, and networking resources. It reduces the need to configure each physical resource manually.

Role of virtualization

- Compute virtualization: Virtual machines share physical servers. New workloads can be deployed quickly without purchasing a separate physical server for every application.
- Storage virtualization: Storage devices are combined into logical pools. Capacity can be allocated, expanded, and managed more easily as demand changes.
- Network virtualization: Virtual switches, software-defined networks, and security policies allow administrators to configure connectivity and segmentation through software.

Traditional datacenter compared with SDDC

Traditional Datacenter	Software-Defined Datacenter
Manual hardware configuration	Software-driven automation
Slower resource provisioning	Faster workload deployment
Separate resource management	Centralized management
Difficult to scale quickly	Flexible resource scaling
More manual maintenance	Policy-based operations

Analysis: By automating provisioning and managing resources through software, an SDDC can respond faster to changing workload requirements. Compute, storage, and network resources can be adjusted without carrying out every change manually on physical devices.

Conclusion: SDDC improves agility through automation, centralized control, flexible scaling, and improved resource utilization. It helps organizations respond quickly to business needs while simplifying datacenter operations.

5. Multi-Cloud Federation: Identity Mismatches and Secure Design

Multi-cloud federation allows users and applications to access services across different cloud providers through trusted identity relationships. Identity mismatches can occur when providers use different user identifiers, role names, authentication protocols, or permission models.

For example, a user who has a limited role in one cloud could accidentally receive excessive privileges in another cloud if the identity and role mappings are configured incorrectly.

Proposed secure identity and privacy design

3. Standardize identity: Use a trusted identity provider (IdP) and a consistent user identifier across cloud environments.
4. Use federation protocols: Implement SAML 2.0 or OpenID Connect (OIDC) to exchange authenticated identity information securely.
5. Map roles carefully: Map external groups and identity claims to local roles using role-based access control (RBAC).
6. Apply least privilege: Give each user only the permissions required for their responsibilities.

7. Protect privacy: Share only necessary identity attributes, encrypt data in transit, and follow applicable data-protection requirements.
8. Strengthen authentication: Enforce multifactor authentication (MFA), validate token signatures and issuers, and use short-lived tokens.
9. Monitor access: Audit sign-ins and role changes, and revoke access when users leave or their permissions change.

Conclusion: A trusted identity provider, standards-based federation, accurate role mapping, and privacy-preserving access policies can resolve identity mismatches while maintaining security across multiple clouds.

6. Equivalent Demonstrations in VMware Workstation Pro

VMware Workstation Pro can be used to demonstrate basic virtualization concepts. These activities are educational simulations of datacenter capabilities, not complete implementations of a production cloud.

6.1 Multiple virtual machines — Multi-instance demonstration

Open VMware Workstation Pro and use the Library pane to display multiple virtual machines. Explain that separate VMs can represent different workloads or instances.

6.2 Snapshots — Version and recovery demonstration

Select a VM and open VM → Snapshot → Take Snapshot. Create a snapshot and open Snapshot Manager to view the saved state. Snapshots are useful for rollback and testing, but they are not a replacement for independent backups.

6.3 Virtual Network Editor — Network virtualization

Open Edit → Virtual Network Editor. Demonstrate NAT, Host-only, and Bridged networking to explain different VM connectivity arrangements.

6.4 Clone — Replica demonstration

Select a VM and choose VM → Manage → Clone. Create a full or linked clone, depending on the available options and requirements. Explain that clones can support testing and demonstrate the idea of creating additional instances.

Practical limitation: VMware Workstation Pro demonstrates local virtualization. By itself, it does not provide full multi-cloud federation, production-scale automatic load balancing, or complete enterprise datacenter orchestration.